

Projections of polyhedral cones

Kent Andersen

In this note we consider a polyhedral cone C , and we study sets obtained from C by projecting out some variables of C . Without loss of generality we assume that it is the last variables of C that are projected out. Let:

$$\text{Proj}_k(C) := \{x \in \mathbb{R}^n : \exists y \in \mathbb{R}^k \text{ such that } \begin{pmatrix} x \\ y \end{pmatrix} \in C\}.$$

be the projection of C onto $\mathbb{R}^n$, where $C \subseteq \mathbb{R}^{n+k}$ is an arbitrary polyhedral cone in $\mathbb{R}^{n+k}$, and $k \geq 1$ denotes the number of variables projected out.

The goal in this note is to prove the following key property of polyhedral cones: Projections of polyhedral cones are also polyhedral cones:

Theorem 1 (*The projection theorem for polyhedral cones*)

The projection of a polyhedral cone onto a subset of the variables is also a polyhedral cone, i.e., the sets $\text{Proj}_k(C)$ for $k \geq 1$ are polyhedral cones. ■

The proof of Theorem 1 uses induction on the number of y -variables. Observe that one has the identity $\text{Proj}_k(C) = \text{Proj}_1(\text{Proj}_{k-1}(C))$:

$$\begin{aligned} x \in \text{Proj}_1(\text{Proj}_{k-1}(C)) &\iff \exists y_1 \in \mathbb{R} \text{ such that } \begin{pmatrix} x \\ y_1 \end{pmatrix} \in \text{Proj}_{k-1}(C) \\ &\iff \exists y_1, \dots, y_k \in \mathbb{R} \text{ such that } \begin{pmatrix} x \\ y \end{pmatrix} \in C. \\ &\iff x \in \text{Proj}_k(C). \end{aligned}$$

The identity $\text{Proj}_k(C) = \text{Proj}_1(\text{Proj}_{k-1}(C))$ gives the inductive step. Indeed, suppose by induction that $\text{Proj}_i(K)$ is a polyhedral cone for all $i < k$ and polyhedral cones $K \subseteq \mathbb{R}^{n+i}$. Hence $\text{Proj}_k(C) = \text{Proj}_1(\text{Proj}_{k-1}(C)) = \text{Proj}_1(Q)$ holds by induction for some polyhedral cone $Q \subseteq \mathbb{R}^{n+1}$. Applying induction again on Q then shows $\text{Proj}_k(C) = \text{Proj}_1(Q)$ is a polyhedral cone.

Hence it suffices to show the basic step, *i.e.*, we must show Theorem 1 holds when $k = 1$. The polyhedral cone C with $k = 1$ can be described with linear inequalities with zero on the right-hand-side (proved in note 1.(B)):

$$C = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^n \times \mathbb{R} : Ax - hy \leq 0_m \right\},$$

where $h \in \mathbb{R}^m$ contains coefficients on the y -variable, and $A \in \mathbb{R}^{m \times n}$ contains coefficients on the x -variables. We use $M := \{1, \dots, m\}$ to index the inequalities $Ax - hy \leq 0_m$. The rows of A are denoted $a_i \in \mathbb{R}^n$ for $i \in M$.

It may not be all inequalities of $Ax - hy \leq 0$ that contain the variable y . Let $M^0 := \{i \in M : h_i = 0\}$ index these inequalities, and let:

$$C^0 = \{x \in \mathbb{R}^n : a_i^T x \leq 0 \text{ for all } i \in M^0\}$$

denote the corresponding polyhedral cone. Every point in $\text{Proj}_1(C)$ satisfies all inequalities $a_i^T x \leq 0$ for $i \in M^0$, so one has $\text{Proj}_1(C) \subseteq C^0$.

If either $h \leq 0_m$ or $h \geq 0_m$, then the inclusion $C^0 \subseteq \text{Proj}_1(C)$ also holds:

Observation 1 *(If all non-zero coordinates have the same sign)*

If either $h \leq 0_m$ or $h \geq 0_m$, then $\text{Proj}_1(C) = C^0$. ■

Indeed, given any $\bar{x} \in C^0$: If $h \geq 0_m$, then $A\bar{x} - hy \leq 0_m$ holds for y sufficiently positive, and if $h \leq 0_m$, then $A\bar{x} - hy \leq 0_m$ holds for y sufficiently negative.

It follows that, if all non-zero coordinates of h have the same sign, then $\text{Proj}_1(C)$ is the polyhedral cone C^0 . In the remainder we therefore assume h has both positive and negative entries. Let:

$$M^+ := \{i \in M : h_i > 0\} \quad \text{and} \quad M^- := \{i \in M : h_i < 0\}$$

index the positive and negative coordinates of h , respectively. The sets M^+ and M^- are both non-empty by assumption.

Observe that the inequalities indexed by $M^+ \cup M^-$ give bounds on the variable y : One has $y \geq \frac{1}{h_i} \cdot a_i^T x$ for $i \in M^+$ and $y \leq \frac{1}{h_i} \cdot a_i^T x$ for $i \in M^-$.

We can now write $\text{Proj}_1(C)$ as a polyhedron. The key step uses a (very) simple observation.

Observation 2 (*A simple observation*)

Let $\alpha_1, \alpha_2 \in \mathbb{R}$. There exists $y \in \mathbb{R}$ such that $\alpha_1 \leq y \leq \alpha_2 \iff \alpha_1 \leq \alpha_2$. ■

We can now write:

$$\text{Proj}_1(C) = \{x \in C^0 : \exists y \in \mathbb{R} \text{ such that:}$$

$$a_{i\cdot}^T x - h_i \cdot y \leq 0 \text{ for all } i \in M^+ \cup M^-\}$$

$$= \{x \in C^0 : \exists y \in \mathbb{R} \text{ such that:}$$

$$\frac{a_{i\cdot}^T x}{h_i} \leq y \leq \frac{a_{k\cdot}^T x}{h_k} \text{ for all } (i, k) \in M^+ \times M^-\}$$

$$= \{x \in C^0 : \frac{a_{i\cdot}^T x}{h_i} \leq \frac{a_{k\cdot}^T x}{h_k} \text{ for all } (i, k) \in M^+ \times M^-\},$$

where the last equality follows from Observation 2. Hence $\text{Proj}_1(C)$ is the solution set for a finite number of linear inequalities, where each inequality has zero on the right-hand-side. Specifically $\text{Proj}_1(C)$ is described by the inequalities $a_{i\cdot}^T x \leq 0$ for $i \in M^0$, and the inequalities $h_i \cdot a_{i\cdot}^T x - h_k \cdot a_{k\cdot}^T x \leq 0$ for $(i, k) \in M^+ \times M^-$.